# The GPU was expensive - Give it something to do!

An Introduction to Shaders in Godot

for Complete Beginners

# WHAT IS A SHADER?

A shader is a small program that uses the GPU to calculate **how each pixel and/or vertex of a given surface should look**.

# GODOT'S MAIN SHADER TYPES

**FRAGMENT SHADERS**

Manipulate Pixel Colors

**VERTEX SHADER**

Manipulate Vertex Positions

# SOME DATA TYPES

| Type | What it is for |
| --- | --- |
| bool, int, uint, float | Known and loved |
| vec2, vec3, vec4 | N component vectors with floats |
| mat2, mat3, mat4 | N component matrices with vectors |

# UNIFORMS

Uniforms are global variables
**that will appear in Godot's sidebar as editable values.**

# SOME BUILT-INS

| Built-In | What it is |
|---|---|
| UV | Pixel position (0..1) across the texture |
| TIME | Seconds since game started |
| COLOR | The pixel's output color (RGBA) |

**UV = (0,0)** is top-left | **UV = (1,1)** is bottom-right

# USEFUL FUNCTIONS

| Function | What it does | Example |
|---|---|---|
| sin(), cos(), tan() | Smooth wave functions | sin(TIME) |
| clamp() | Restrict value between min/max | clamp(val, 0.0, 1.0) |
| mix() | Blend between two values | mix(a, b, t) |
| smoothstep() | Soft edge transition | smoothstep(0.3, 0.7, x) |
| abs(), floor() | Absolute value / round down | abs(sin(TIME)) |
| texture() | Used to read the sprite2d texture | texture(TEXTURE, UV) |

# DEMO TIME!

# WHAT WE COVERED

1. What a Shader Is
2. Godot's Shader Types
3. Data Types
4. Uniforms
5. Built-in Variables
6. Useful Functions

# MATERIAL

https://github.com/elkdisk/godot-shader-talk-2026